N084AV (S/N: W20041874922) — Primary Storage Failure Report

Immfly Ground Operations / IT Session: September 21, 2026

Field	Value
Machine	**N084AV** (S/N: W20041874922)
Affected component	Primary Storage Drive (`hd0` / Solid State Drive) — Boot sector failure
Initial symptom	GRUB bootloader physical read abort: `error: failure reading sector 0x79aee0 from 'hd0'`, followed immediately by fatal Linux Kernel Panic (`unable to mount root fs on unknown-block(0,0)`).
Hardware status	HARDWARE FAILURE — Physical media bad sector on storage unit `hd0` at sector `0x79aee0` (LBA 7,974,624). Unrecoverable ECC failure.
Root cause	Permanent NAND flash memory cell corruption within the boot partition. The bootloader failed to load the initial ramdisk (`initrd`) image. As a consequence, the Linux kernel boots without required storage and filesystem drivers, preventing partition table resolution and triggering a fatal root mount panic.
Comparison	Distinct from auxiliary partition mount failures (N067AV, which booted into Emergency Shell) and peripheral modem disconnects (N109AV). On N084AV, the primary storage sector failure strikes early boot artifacts on `hd0`, preventing the operating system from completing bootstrap initialization.
Recommended action	**Physical primary storage drive (SSD) replacement required.** Complete re-imaging of operating system and media content deployment.

NARRATIVE SUMMARY

During boot sequence initialization on aircraft server unit **N084AV (S/N: W20041874922)**, the system experienced a fatal storage hardware failure at the earliest bootloader stage. The GRUB stage-2 reader encountered an unrecoverable medium read fault on sector `0x79aee0` of the primary disk drive (`hd0`), producing the alert `error: failure reading sector 0x79aee0 from 'hd0'`.

Sector `0x79aee0` (LBA 7,974,624) corresponds to the storage block range where the kernel's initial ramdisk image (`initrd.img-5.4.0-216-generic`) is hosted. Because the storage controller returned an uncorrectable ECC read error to the BIOS/bootloader, the system skipped loading the initrd payload into RAM and invoked the bare kernel directly with root parameter `UUID=c92b5677-bd29-427c-98ac-207b10602933`.

Without the storage modules, AHCI/NVMe drivers, and block layer device mappers supplied by the ramdisk, the Linux kernel was unable to discover or enumerate any attached block devices. As documented in the console trace, the kernel's available partition table was completely empty, culminating in `VFS: Cannot open root device ... or unknown-block(0,0): error -6` and an unrecoverable **Kernel Panic** (`CPU: 0 PID: 1 Comm: swapper/0`). The machine is completely out of service and cannot boot into multi-user mode.

N084AV (S/N: W20041874922) — Diagnostic Timeline & Console Appendix

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DIAGNOSTIC FINDINGS & SEQUENCE OF EVENTS

1. **Bootloader Read Failure (GRUB):**
   During pre-boot payload staging, GRUB attempted to read sector `0x79aee0` from primary drive `hd0`. The drive controller signaled a persistent read fault, halting automated execution and requiring user keystroke bypass.
2. **Kernel Image Decompression & Initialization:**
   The raw kernel executable (`vmlinuz-5.4.0-216-generic`) loaded into physical memory. Early hardware probes enumerated CPU cores, ACPI tables, and PCIe root complexes on host platform `SecureGUARD SGP100/SG-S283`.
3. **Missing Initial Ramdisk (initrd):**
   Due to the earlier read error on sector `0x79aee0`, the initrd archive was corrupted/unloaded. The kernel proceeded with built-in modules only, omitting user-space root pivot scripts and dynamic device rules.
4. **Block Device Discovery Failure:**
   At timestamp `4.459870s`, the Virtual File System (VFS) layer attempted to mount root device `UUID=c92b5677-bd29-427c-98ac-207b10602933`. Lacking driver access to the underlying storage partitions, the block layer returned `unknown-block(0,0): error -6`.
5. **Fatal Kernel Panic:**
   With zero storage partitions available, the kernel triggered `panic()` in `mount_block_root+0x23f`, terminating process `PID: 1 (swapper/0)` and rendering the server completely inoperative.

CONCLUSION AND RECOMMENDATION

The primary storage unit (`hd0`) in server **N084AV** has sustained permanent physical hardware degradation. Defective NAND flash sectors within critical boot paths prevent system bootloader loading, and software-based recovery is impossible due to hardware I/O rejection at the controller level.

Recommended action: Issue an immediate Line Maintenance Work Order to **physically replace the primary storage drive (SSD)** in server unit **N084AV (S/N: W20041874922)**, followed by operating system base image deployment and media synchronization.

APPENDIX: CONSOLE BOOT FAILURE & PANIC OUTPUT

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[BOOT:N084AV] GRUB sector read failure and fatal Kernel Panic stack trace:
error: failure reading sector 0x79aee0 from `hd0'.

Press any key to continue...
[    0.000000] Linux version 5.4.0-216-generic (buildd@lcy02-amd64-014) #236-Ubuntu SMP
[    0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-5.4.0-216-generic root=UUID=c92b5677-bd29-427c-98a
c207b10602933 ro console=ttyS0,115200n8
[    0.000000] DMI: SecureGUARD SGP100/SG-S283, BIOS ES283ISG.115 05/12/2021
[    1.713919] smpboot: CPU0: Intel(R) Atom(TM) CPU C3758R @ 2.40GHz (family: 0x6, model: 0x5f, stepping:
0x1)
[    2.241640] pci 0000:02:00.0: [1cc1:8201] type 00 class 0x010802
[    4.436927] md: Waiting for all devices to be available before autodetect
[    4.453693] md: autorun ...
[    4.456499] md: ... autorun DONE.
[    4.459870] VFS: Cannot open root device "UUID=c92b5677-bd29-427c-98ac-207b10602933" or unknown-block
(0,0): error -6
[    4.470393] Please append a correct "root=" boot option; here are the available partitions:
[    4.473750] Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0)
```

```
[    4.478750] Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0)
[    4.493527] CPU: 0 PID: 1 Comm: swapper/0 Not tainted 5.4.0-216-generic #236-Ubuntu
[    4.501179] Hardware name: SecureGUARD SGP100/SG-S283, BIOS ES283ISG.115 05/12/2021
[    4.508831] Call Trace:
[    4.511291]  dump_stack+0x6d/0x8b
[    4.514614]  panic+0x114/0x2fd
[    4.517675]  mount_block_root+0x23f/0x2e8
[    4.521685]  mount_root+0x38/0x3a
[    4.525004]  prepare_namespace+0x13f/0x194
[    4.529103]  kernel_init_freeable+0x265/0x289
[    4.533462]  ? rest_init+0xb0/0xb0
[    4.536867]  kernel_init+0xe/0x110
[    4.540275]  ret_from_fork+0x35/0x40
[    4.558115] ---[ end Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,
0) ]---
```